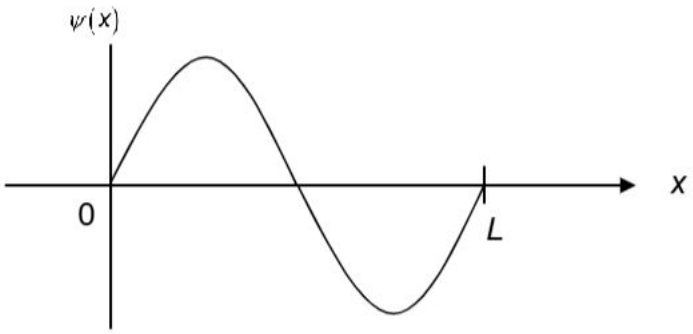
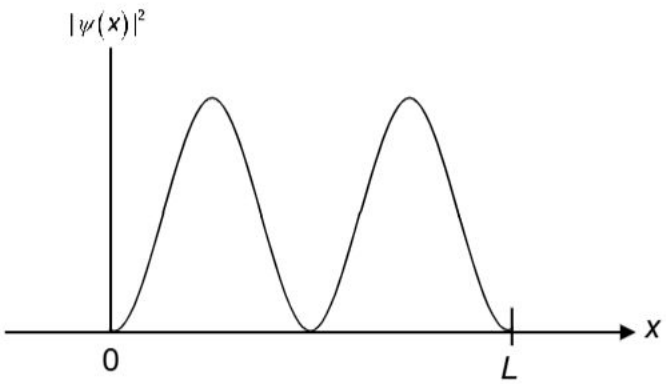
Question 7		
(a)(i)	 Or negative sine.	B1
(a)(ii)		B1
(b)(i)	A normalised wave function is one where the probability of finding the particle in all space / across all x equals 1 Or where the total area under the probability density function equals 1.	B1
(b)(ii)	$\frac{A^2}{2} \times L = 1$ $A = \sqrt{\frac{2}{L}}$	B1
(c)	Energy of photon emitted $= \frac{(2)^2 h^2}{8mL^2} - \frac{h^2}{8mL^2} = \frac{3h^2}{8mL^2}$ $= \frac{3\left(6.63 \times 10^{-34}\right)^2}{8\left(0.15 \times 9.11 \times 10^{-31}\right)\left(2.2 \times 10^{-9}\right)^2}$ $= 2.4923 \times 10^{-19}$ J Wavelength of photon emitted $= \frac{hc}{E}$ $= \frac{\left(6.63 \times 10^{-34}\right)\left(3.00 \times 10^{8}\right)}{2.4923 \times 10^{-19}}$ $= 7.98 \times 10^{-7}$ m	C1 C1 A1